

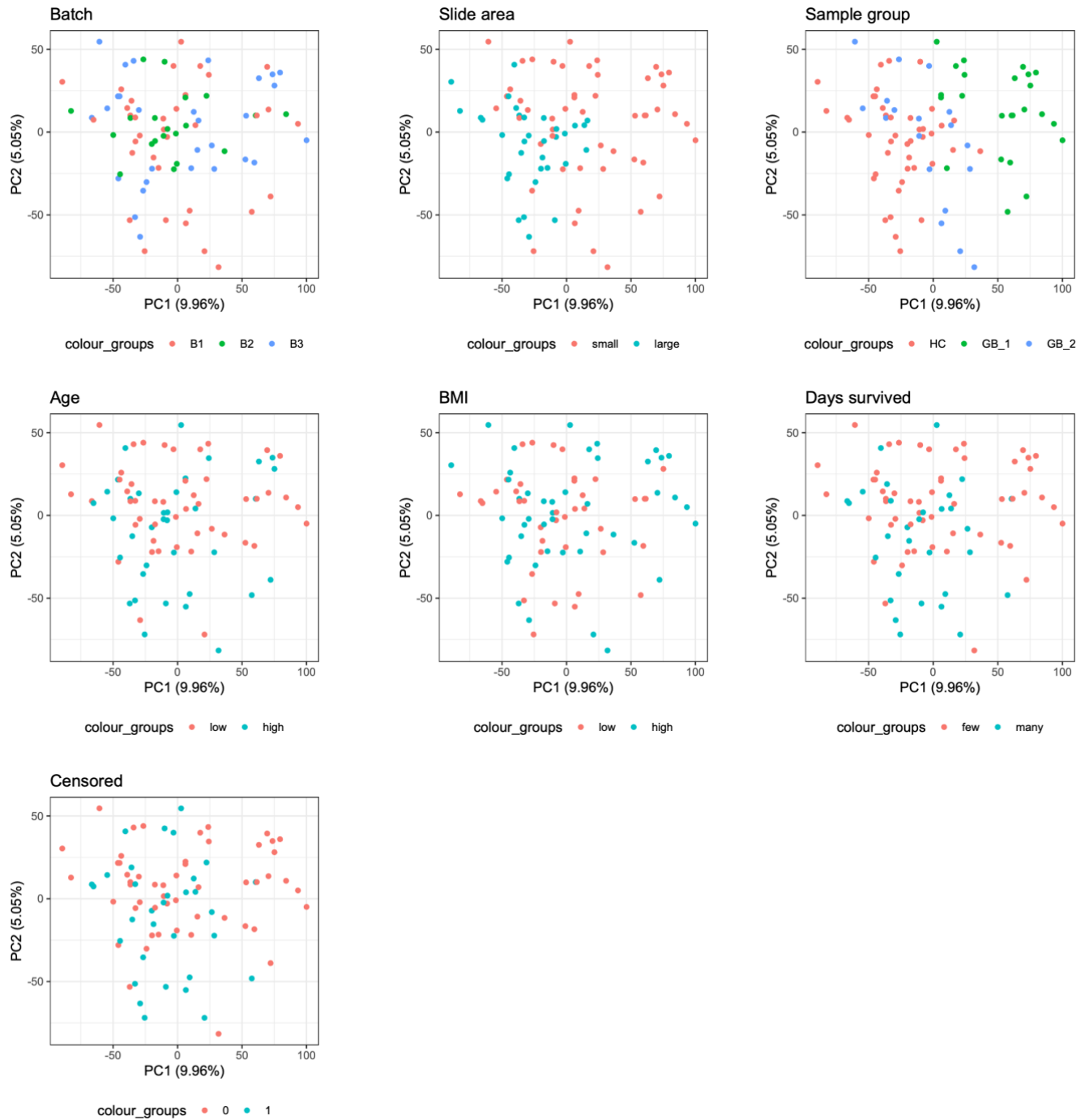
Table. Clinical and technical characteristics of patients within sample groups

Characteristic	Healthy Controls (HC) N = 42 ¹	Glioblastoma Metastatic (GB_1) N = 24 ¹	Glioblastoma Non-metastatic (GB_2) N = 18 ¹	P value ²
Clinical				
Age	48(20)	48(19)	48(22)	>0.9
BMI	25.6(6.4)	26.4(6.2)	24.5(6.7)	0.7
Censored				0.004
0	26/42(62%)	20/24(83%)	6/18(33%)	
1	16/42(38%)	4/24(17%)	12/18(67%)	
Days_survived	906(301)	739(198)	1,129(270)	<0.001
Technical				
Slide_area	127,168(32,988)	51,976(21,962)	48,716(25,462)	<0.001
Batch				0.8
B1	17/42(40%)	10/24(42%)	8/18(44%)	
B2	12/42(29%)	4/24(17%)	4/18(22%)	
B3	13/42(31%)	10/24(42%)	6/18(33%)	
¹ Mean (SD); n/N(%)				
² Kruskal-Wallis rank sum test; Fisher's exact test; Pearson's Chi-squared test				

Clinical and technical characteristic table interpretation

This experiment has been well designed based on the distribution of samples across the three batches. Each batch includes all three sample groups, and there are no significant differences between batches within the groups ($p=0.8$). In addition, there are no significant differences in age or BMI among the patients, minimising potential confounding effects ($p>0.7$). However, the average tissue size used for sequencing in the control samples was considerably larger than that used for the glioblastoma samples, producing a source of data variation. On the other hand, patients with metastasised glioblastoma had less days of survival when compared to non-metastasized glioblastoma patients ($p<0.001$). Unexpectedly, the glioblastoma-metastasised group showed a low number of death events despite having less days of survival. This inconsistency may reflect errors in data recording, such as misclassification of the censoring variable.

Principal component analysis before correction



In this experiment, there are two potential technical cofounders: batch and slide area. The PCA coloured by batch shows no clear separation of samples according to batch. This observation is consistent with the summary table, which indicated that the sample groups are evenly distributed

across the three batches. Therefore, batch does not appear to introduce a significant technical effect, and correction for batch is not required. Additionally, PCA's colored by clinical characteristic: age and BMI look not to influence the biological condition. However, when the PCA plot is colored by slide area, the sample groups appear to separate according to slide area position, showing that slide area may be confounding the biological conditions. This observation is consistent with the summary table, which illustrated the significant difference of slide area in health control samples and glioblastoma samples. Therefore, indicating the need for correction by slide area.